

Week 6 — Wiring Diagrams & Schematics

Sessions: 1 (Theory) · 2 (Lab) · **Total:** 4 hours

Primary LOs: LO-08

Hands-on target: Session 2 ≈ 85% practical

Instructor Preparation Guide (Complete Day Before)

Pre-session setup

- [] Print or display **correct model-year** diagrams for each training cart (Club Car, EZ-GO, Yamaha)
- [] Large schematic on screen/poster for group tracing
- [] Prepare planted faults: open, short-to-ground training lead, high-resistance crimp
- [] Crimping station: ratcheting crimper, terminals, heat shrink, dielectric grease, soldering iron
- [] Print labs/W06_harness_fault_isolation.md and handout wiring_symbol_glossary.md
- [] Fuse/relay assortment for testing practice

Planted faults (label carts after lab for reset)

Fault type	Safe planting method
Open	Inline training fuse removed / connector pin backed out (logged)
High resistance	Intentionally poor training splice in non-critical lamp circuit preferred; or documented high-R jumper
Short	Use fused short box on low-energy circuit — never uncontrolled pack short

Tools

DMM, wiring diagrams, pin probes, mirror, contact cleaner, crimp tools.

Week 6 Learning Objectives

1. Read and navigate OEM golf cart wiring diagrams.
 2. Identify standard schematic symbols; distinguish power vs signal circuits.
 3. Trace systematically to locate opens, shorts, and high-resistance connections.
 4. Perform basic connector repair to shop standard.
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Session 1 — Theory & Demonstration (2 hours)

Materials needed: handouts/wiring_symbol_glossary.md; large schematic display; printed model-year diagrams for demo cart; bad-crimp vs good-crimp samples; voltage-drop demo leads; whiteboard.

Timed Agenda

Time	Min	Activity	Instructor Notes
0:00	5	Safety: probing live circuits; one-hand rule	
0:05	15	Symbol literacy drill	Use glossary handout; [DIAGRAM PLACEHOLDER: symbol sheet poster]
0:20	15	Power vs signal: cable gauge & expectations	Thick pack cables vs throttle sense wires
0:35	15	Brand harness notes: color codes vary — verify diagram	Club Car vs EZ-GO vs Yamaha examples
0:50	5	Break	
0:55	20	Guided group trace: battery+ → solenoid → controller → motor	Everyone points on printed diagram
1:15	15	Common failure points: chafe, water, rodents, bad grounds	Photos
1:30	20	Method: voltage drop vs continuity for high R	Teach when each is valid
1:50	10	Lab preview; wrong-year diagram caution story	

Key Teaching Points

- Wrong model-year diagram is worse than no diagram.
- Continuity with power disconnected; voltage drop with controlled power applied.
- Grounds are suspects — measure them.
- Dielectric grease after proper clean/crimp — not a substitute for metal-to-metal contact.

Common Misconceptions

Misconception	Correction
“Same color wire means same function across brands.”	Colors are OEM-specific.
“Beep = good cable for 200A.”	Use voltage drop under load for power cables.
“Twisting wires is a fine repair.”	Use proper terminals, crimp height, strain relief, seal.

Safety Reminders

- Pull fuses / isolate before cutting or splicing.
- Protect circuits with proper fuse rating after repair — never copper-slug.
- Soldering irons: burn/fire awareness; stand required.

Session 2 — Hands-On Lab (2 hours)

Materials needed: labs/W06_harness_fault_isolation.md; planted open/high-R faults (fused short box if used); crimp station consumables; DMM; dielectric grease; heat shrink; matching diagrams.

Timed Agenda

Time	Min	Activity	Instructor Notes
0:00	5	Assign carts + diagrams; PPE	Confirm diagram matches cart tag
0:05	20	Trace main power path on cart while marking diagram	Checkpoint with instructor
0:25	20	Test key switch, direction, throttle enable, solenoid coil feed	Record findings
0:45	5	Break	
0:50	30	Locate planted fault using DMM + diagram	Complete isolation form
1:20	25	Connector repair practice: cut, strip, crimp, seal	Instructor QC crimps
1:45	15	Debrief faults; restore training circuits; cleanup	

Assessment Focus

Diagram navigation quiz + successful location/repair of planted wiring fault.

Week 6 Quiz (8 questions)

1. Before tracing a cart, you must confirm the diagram matches:
 - A) Only paint color
 - B) Make, model, and year/configuration
 - C) Tire brand
 - D) Owner's favorite radio station
2. A solenoid coil circuit is typically a:
 - A) Very high current traction cable
 - B) Control/signal-level circuit relative to motor cables

- C) Fuel line
 - D) Shock absorber
3. Best method to evaluate a main battery cable under load:
- A) Smell test
 - B) Voltage drop measurement while current flows
 - C) Paint thickness
 - D) Horn volume
4. Standard schematic symbols should be learned because:
- A) They are identical to tire codes
 - B) They allow rapid identification of components across diagrams
 - C) They replace OEM manuals entirely
 - D) They are unused in golf carts
5. True or False: Water intrusion at connectors can cause intermittent high resistance and corrosion.

Short Answer

1. List three common physical harness failure points.
2. Describe the difference between an open and a short.
3. Why apply dielectric grease after a quality crimp on exposed connectors?

Answer Key

1. **B**
2. **B**
3. **B**
4. **B**
5. **True**
6. Chafe points, grommets, under-seat moisture zones, rodent areas, ground lugs, etc.
7. Open = broken path / infinite resistance; short = unintended low-resistance path to another circuit/ground
8. Moisture exclusion / corrosion slow-down — does not fix a bad crimp